

Credit Performance Diagnostic & Scale Reassessment

Optimizing the Profitability of an Already Highly Lucrative Portfolio (\$3.33B in Net Profit Generated) and Mapping up to \$299M in Opportunity Cost to Be Recovered

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SCOPE SQL & Power BI Pipeline • 1.34 Million Closed Contracts • \$20.3 Billion in Capital Funded (2007–2018)

SOURCE Kaggle — Lending Club Loan Data (2007–2018Q4), accepted_2007_to_2018Q4.csv

\$3.33B

Accumulated Net Income

\$20.3B

Total Volume Funded

1.34M

Closed Credit Cycles

4.43%

Current Consolidated ROA

An already highly profitable portfolio — the analysis ahead reveals where to unlock even more value at scale.

* Monetary Note: All financial figures in this study were inflation-adjusted via CPI to Constant 2018 Dollars, neutralizing distortions between the 2007 and 2018 vintages.

The Business Model & Credit Marketplace

Disruption in Unsecured Personal Credit (Unsecured Consumer Loans)



Marketplace Model

Digital disintermediation and operational efficiency: 100% digital origination, with no physical branches or traditional teller infrastructure.



The Product

Fixed-rate personal loans over 36 and 60 months, focused primarily on consolidating credit card debt and other high-rate liabilities.



Underwriting

Risk grade assignment (Grades A through G) via a proprietary Risk-Based Pricing algorithm, calibrating interest rates to default probability.

1.34 million applications approved and funded over the period analyzed (2007–2018) — a scale that reflects the maturity and reach of the marketplace model.

Business Robustness and the Case for Superior ROA

How an Already Highly Profitable Model Outperforms Market Benchmarks

A Profitable Model by Design

With \$3.33B in net profit and a 4.43% ROA, the bank already outperforms the market. Even so, without real collateral, returns must embed a risk premium to absorb expected delinquency.

Absence of Collateral

- Without real collateral, recovery upon default depends almost exclusively on collections, raising the expected loss per unit of credit.

Required Risk Premium

- The institutional Hurdle Rate of 4.00% p.a. defines the minimum ROA floor for origination to remain economically sustainable.

ROA Benchmark Table (2018)

Reference	ROA (p.a.)
US Commercial Banks (FRED Average, 2018)	~1.00%–1.32%
Discover Financial (DFS)	2.70%
Synchrony Financial (SYF)	3.80%
Happen Bank Institutional Hurdle (Target)	4.00%
Happen Bank Historical Result (Actual)	4.43%

Consolidated result: \$3.33B in net profit on \$20.3B funded (1.34M contracts).

* Methodological Note (ROA & Time-Aligned Benchmarking): Return on Assets (ROA) measures the portfolio's net profitability (net financial result / capital funded). To ensure analytical consistency, market benchmarks (FRED, Discover Financial, and Synchrony Financial) were consolidated from official 2018 financial reports and macroeconomic data, aligning rigorously with the dataset's historical window close (2007–2018) against the institutional Hurdle Rate of 4.00% p.a.

Analytical Architecture: Segmentation & Feature Engineering

Proprietary Lenses for Fine-Grained Risk and Profitability Diagnostics, with Full Volumetrics

Data-Driven Income Bands (USD 2018)

A	Class A — Above \$104,000
	268,487 contracts
B	Class B — \$77,000 to \$104,000
	269,294 contracts
C	Class C — \$59,800 to \$77,000
	267,598 contracts
D	Class D — \$44,100 to \$59,800
	268,295 contracts
E	Class E — Up to \$44,100
	271,315 contracts

Budget Commitment (Installment-to-Income)

Light

- ~965K contracts (estimated) · \$12.58B funded (Real Values) — Below 10% of gross monthly income committed to the installment.

Moderate — "The Middle"

- 367,023 contracts · \$7.40B funded — 10% to 20% of income committed, the highest-volume band and, as we'll see, the one with the greatest hidden risk.

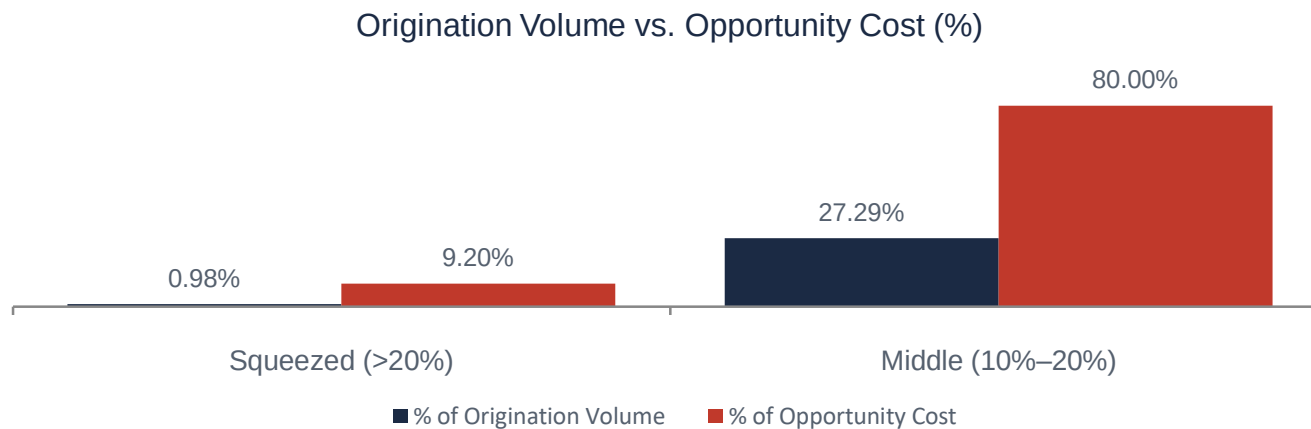
Severe ("Squeezed")

- 13,163 contracts · \$319M funded — Above 20% of income committed, the portfolio's intuitively riskiest profile.

** Methodological Note (Feature Engineering & SQL): Income classes were calibrated via quantiles over annual income adjusted for CPI (2018 Dollars); the distribution results in ~268K contracts per class. | * Calibration vs. Market DTI: While the traditional DTI measures stress above 30%–35% of total budget, this metric measures the individual product installment's commitment — in unsecured credit, above 20% the financial stress becomes critical, justifying the "Squeezed" classification through the empirical ROA collapse observed in the data.*

The Silent Danger of the Portfolio's "Middle"

The \$5.02B Volume in the Middle vs. The Myth of the "Squeezed" Customer



THE MYTH OF THE "SQUEEZED"

13,163 contracts • 0.98% of the base

\$319M funded • \$27.6M in opportunity cost

THE DRAIN OF "THE MIDDLE"

367,023 contracts • 27.29% of sales

\$7.40B funded • \$239.2M in cost (80% of total)

Scale is the real risk: the "Squeezed" segment holds just 0.98% of contracts, but the "Middle" represents 27.29% of the base — 28 times more volume. Each percentage point of ROA below the hurdle in this band multiplies across a 28x larger scale, turning an apparently moderate risk into the bank's biggest cash drain: \$239.2M.

* Methodological Note (Opportunity Cost & Effective Duration): Opportunity Cost (\$ USD) is the difference between the Hurdle Rate (4.00% p.a.) and the actual ROA of each cell, multiplied by the funded volume and adjusted for the contracts' Effective Duration (weighted average horizon of 3.76 years) — converting the annualized percentage gap into accumulated financial value over the loans' real lifecycle (2007–2018).

Breakdown by Credit Grade in the Middle

Grades C, D, and E Concentrate \$185.7M (78%) of the Middle's Opportunity Cost

GRADE C	GRADE D	GRADE E
CONTRACTS 105,652	CONTRACTS 74,151	CONTRACTS 42,252
FUNDED VOLUME \$2.07B	FUNDED VOLUME \$1.50B	FUNDED VOLUME \$913M
ACTUAL ROA (p.a.) 3.58%	ACTUAL ROA (p.a.) 2.76%	ACTUAL ROA (p.a.) 2.40%
OPPORTUNITY COST \$42.1M	OPPORTUNITY COST \$78.6M	OPPORTUNITY COST \$65.0M
Other Grades (A/B, F/G): 144,968 contracts · \$2.93B funded · \$53.5M in cost — Total in the Middle: 367,023 contracts · \$7.40B funded · \$239.2M in opportunity cost (78% concentrated in Grades C, D, and E).		

Complete Performance View by Credit Grade

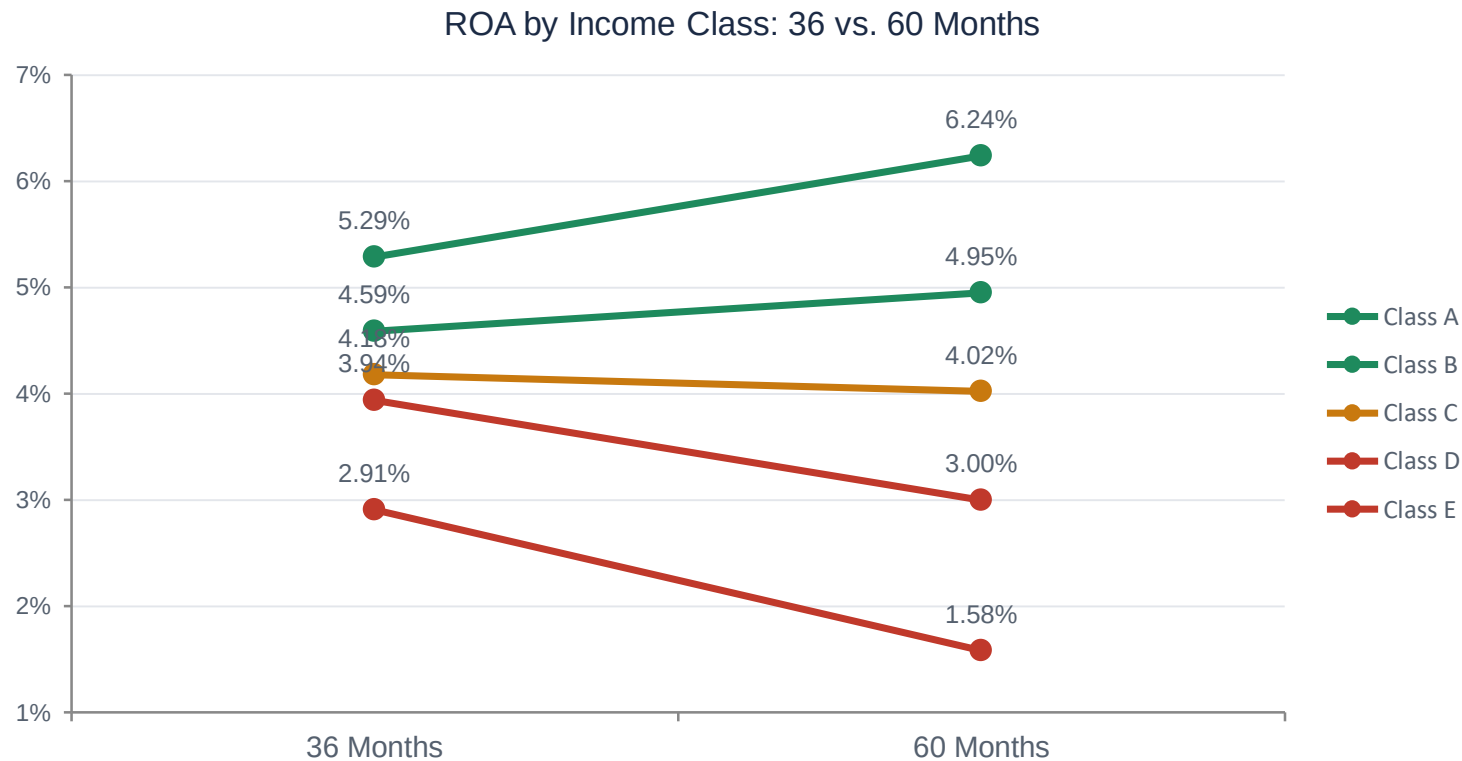
Contracts, Funded Volume, Actual ROA, and Opportunity Cost — All Bands

Credit Grade	Commitment	Contracts	Funded Volume	Actual ROA (p.a.)	Opportunity Cost
Grades A/B	10%–20%	121,844	\$2.40B	4.63%	\$4.7M
Grade C	10%–20% (Middle)	105,652	\$2.07B	3.58%	\$42.1M
Grade D	10%–20% (Middle)	74,151	\$1.50B	2.76%	\$78.6M
Grade E	10%–20% (Middle)	42,252	\$913M	2.40%	\$65.0M
Grades F/G	10%–20%	23,124	\$524.5M	1.98%	\$48.8M
Subtotal Middle	10%–20%	367,023	\$7.40B	3.38%	\$239.2M (80%)
Squeezed	Above 20%	13,163	\$319M	1.37%	\$27.6M (9%)
Total Bank	All Bands	1,344,989	\$20.30B	4.43%	\$299.5M

The Middle (10%–20%) accounts for 27.29% of contracts and 80% of the bank's entire opportunity cost — the clearest evidence of hidden risk at scale.

Term Elasticity and the 60-Month Trap

How the 5-Year Duration Boosts the Upper Class but Collapses the Base of the Pyramid



CONCEPT: TERM FATIGUE

In fragile income profiles, accumulated risk over 5 years is not linear — delinquency among the most vulnerable households accelerates and outpaces the additional interest revenue generated by the extended term.

THE DIVERGENCE

Classes A and B gain from the extended term. Classes D and E collapse. The full breakdown, with volumes and \$ impact, is on the next page.

** Methodological Note (Credit Term Fatigue): The term elasticity analysis demonstrates the divergence in the expected loss (EL) curve over duration. In fragile income profiles (Classes D and E), accumulated risk is not linear but exponential beyond the 36th month, outpacing the interest revenue generated by extending the term.*

Volume and Financial Impact by Income Class

Contracts, Funded Volume, and Nominal Result at Each Term

Class	Contracts	Funded Volume	ROA 36m	ROA 60m	Δ (pp)	Nominal Impact
Class A	268,487	\$4.21B	5.29%	6.24%	++0.95 pp	+\$96.0M
Class B	269,294	\$3.95B	4.59%	4.95%	++0.36 pp	+\$32.2M
Class C	267,598	\$3.68B	4.18%	4.02%	-0.16 pp	Neutral (-\$11.9M)
Class D	268,295	\$3.41B	3.94%	3.00%	-0.94 pp	-\$51.4M
Class E	271,315	\$3.15B	2.91%	1.58%	-1.33 pp	-\$34.2M

The average hides the risk: the aggregate impact of extending to 60 months is +\$30.7M net overall — but that average masks a +\$128.2M gain in Classes A and B and a -\$85.6M loss in Classes D and E.

The Pricing Paradox in Grade G

Analysis of \$196.4M Funded: Why the 60-Month Term Saves the Bank's Worst Credit Grade

Total Grade G Volume: 9,126 contracts • \$196.4M funded

GRADE G — 36 MONTHS

-1.52%

ROA — Net Loss

1,518 contracts • \$23.1M funded

Nominal Loss: **-\$1.05M**

At 36 months, accumulated interest revenue still doesn't offset the initial defaults concentrated in the bank's worst risk class.



GRADE G — 60 MONTHS

+3.76%

ROA — Net Profit

7,608 contracts • \$173.3M funded

Nominal Profit: **+\$32.59M**

At 60 months, the gross interest margin from performing borrowers over the extended term absorbs the defaults and generates consistent net profit.

Strategic Action Plan & Committee Recommendations

Three Immediate Interventions with Estimated Financial Gain



1. Surgical Term Lock

Block the 60-month term for Classes D and E, redirecting 100% of origination flow to the 36-month term.

**Avoids up to \$85.6M in losses
(Classes D and E)**



2. Repricing the Middle

Adjust +45 to +80 bps on Grades C, D, and E within the 10%–20% income-commitment band.

**Direct recovery of up to \$123.75M in
Grades C, D, and E**



3. Reallocating Grade G

Eliminate the 36-month term for Grade G, channeling 100% of origination to the 60-month term.

**Reverses -\$1.05M and adds +\$4.3M in
profit**

Coverage of this phase: \$123.75M of \$299.5M diagnosed. The remaining \$175.75M — concentrated in specific subpopulations within Grades A/B, F/G, and the Choked segment — is the subject of the next analysis phase, with sub-grade segmented repricing and possible underwriting review.

CONCLUSION

Conclusion & Unlocked Value Projection

Turning Data Intelligence into Profitable Market Leadership

4.43% → 4.57%

Projected Increase in Consolidated Portfolio ROA

\$123.75M

Opportunity Cost Recovered — Middle Repricing (Phase 1)

CONSOLIDATING HIGH-YIELD ROA LEADERSHIP



SCALE ALREADY CONSOLIDATED

\$20.3B • 1.34M contracts

\$3.33B in net profit already generated

Data intelligence applied to risk-based pricing turns an already profitable portfolio (\$3.33B in results) into an even more efficient cash-generating machine. In this first phase, \$123.75M of the \$299.5M diagnosed already has an approved action plan; the remaining \$175.75M moves to the next phase of sub-population repricing.

** Methodological Note (Governance & Simulation): Incremental ROA projections assume the maintenance of application attrition rates and demand elasticity from the 2007–2018 cycle, assuming neutrality from external macroeconomic shocks.*